Retail Demand Forecasting & Inventory Optimization

A Data Analytics Portfolio by Yujin Jeon

Executive Summary: Predictive precision drives direct business impact.

10M+

Analyzed retail transactions (2020-2023)

HistGB

Tree-based ML model optimized for non-linear patterns

-21.8%

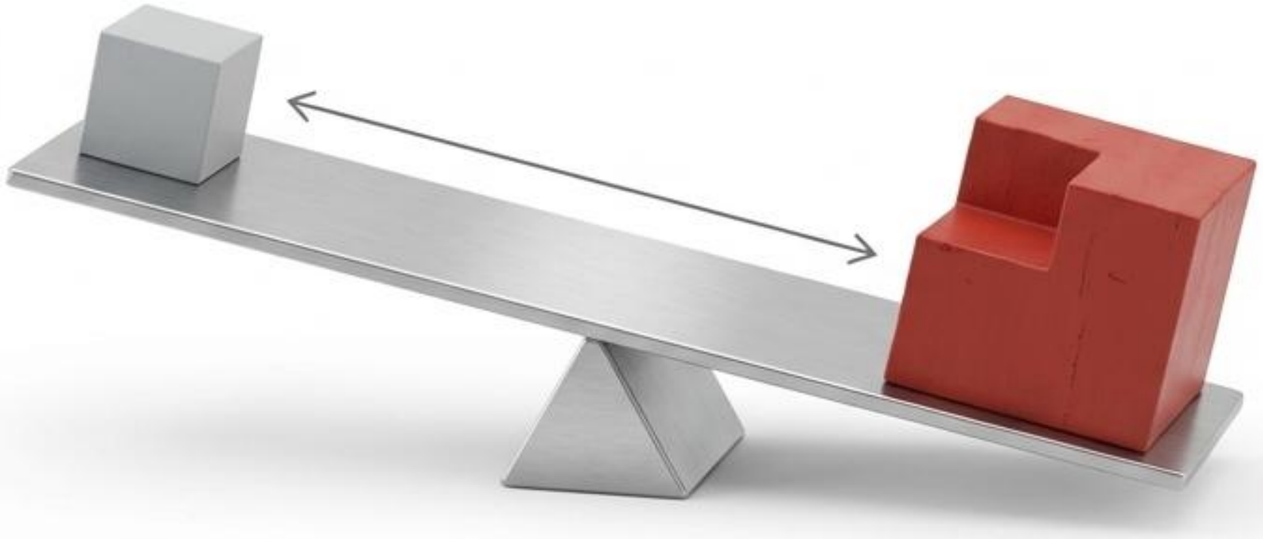
Forecast error reduction compared to traditional statistical baseline

3,329 Units

Inventory misallocation prevented over a 90-day evaluation

Business Problem:
The asymmetry of retail risk.

Overstock
Holding Cost: 2 / unit



Stockout
Lost Sales: 8 / unit

Goal: Move beyond simple seasonal forecasting. Leverage feature-based ML to find the exact optimal point of cost balance.

Data Engineering Blueprint: Scaling to 10M+ rows.

10M+ retail transactions (2020–2023) · daily product-category sales data

1. Chunking

500,000 row batch processing to completely prevent memory overflow.

→

2. Cleansing

Missing value imputation and category/product string normalization.

→

3. Feature Engineering

Key engineered features: weekday seasonality, hourly demand patterns, rolling demand volatility, recent sales momentum

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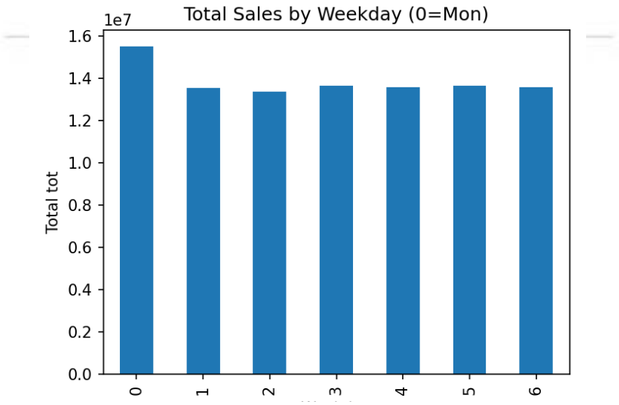
4. Parquet Storage

Compressed storage to maximize I/O speed.

Statistical Validation:
Grounding features in mathematical reality.

H1: Day of Week Effect

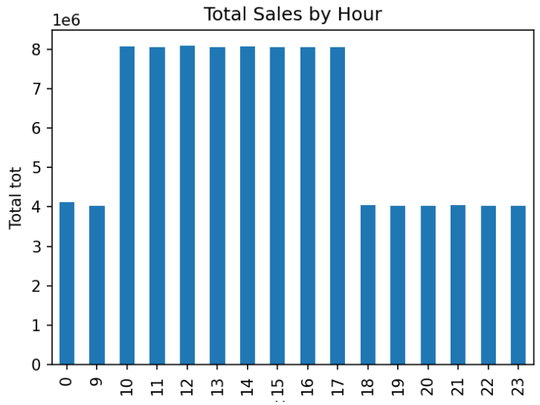


ANOVA Test

p-value = 0.00

Weekday feature confirmed

H2: Time of Day Effect

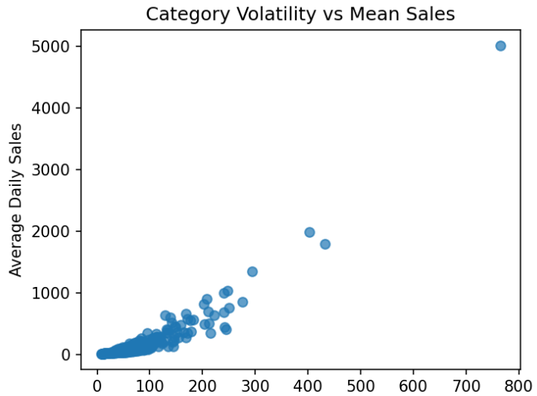


ANOVA Test

p-value = 0.00

Hour feature confirmed

H3: Demand Volatility Correlation



Pearson Corr.

r = 0.908

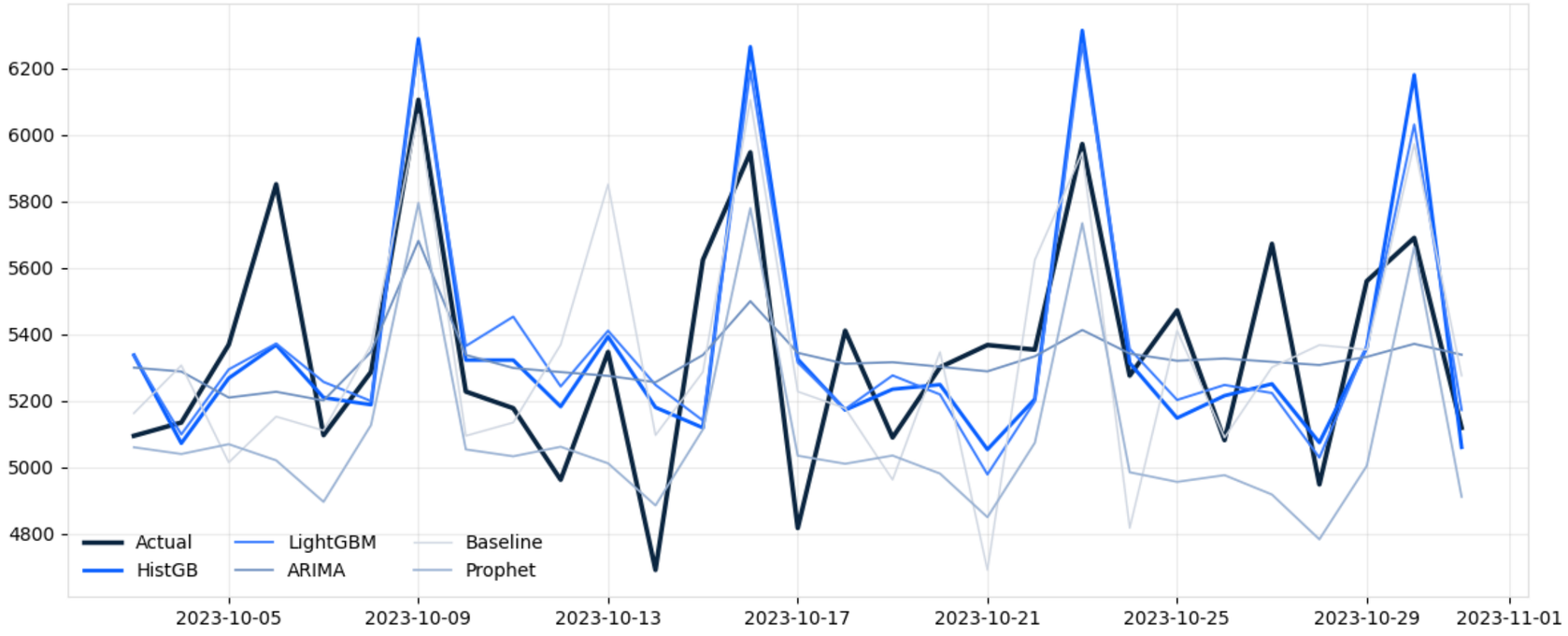
Positive corr. validates non-linear tree modeling

Model Benchmark: HistGB outperforms traditional time-series.

Model	RMSE	WMAPE
Seasonal Naive (Baseline)	394.69	5.61%
Prophet	372.29	5.71%
ARIMA	355.66	5.61%
LightGBM	313.11	4.99%
HistGB (Winner)	**308.75**	**4.91%**

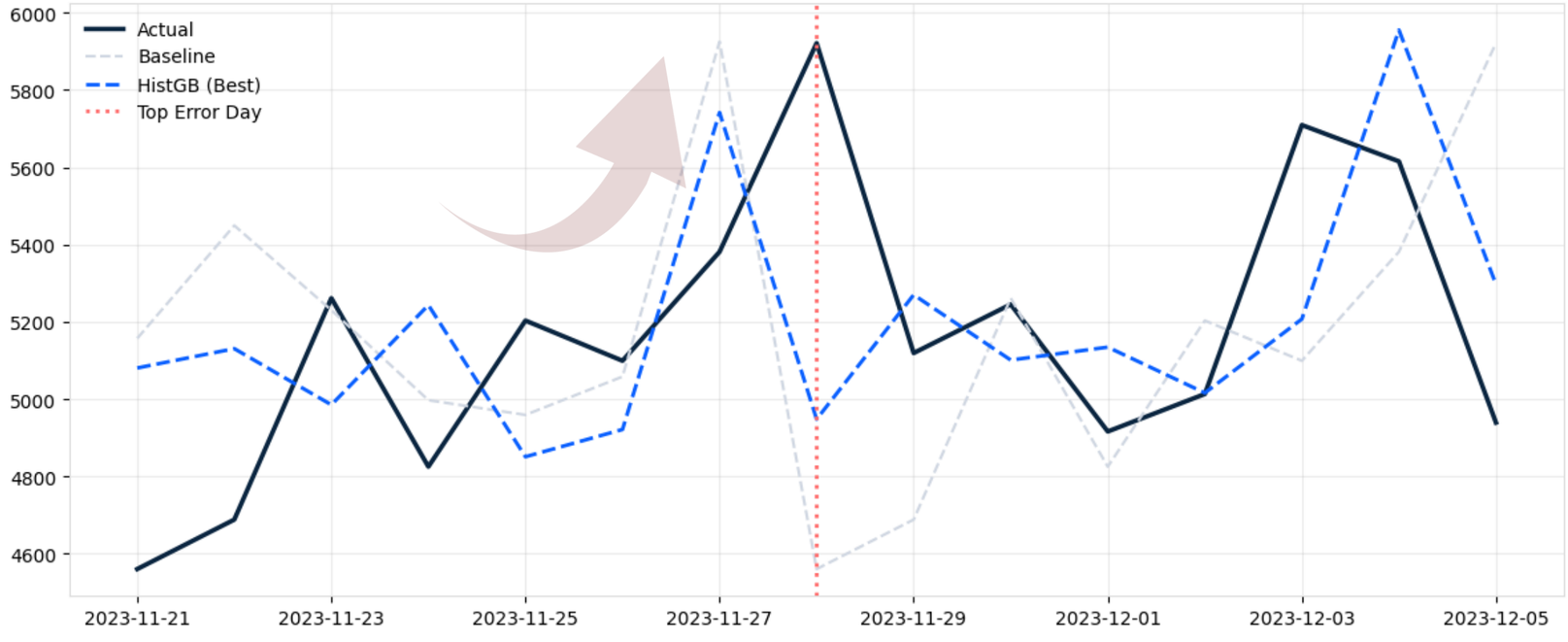
-21.8% Error Reduction vs. Baseline

Forecast Performance: Tracking reality with precision.



ML-based **HistGB** consistently tracks volatile demand patterns that traditional time-series completely miss.

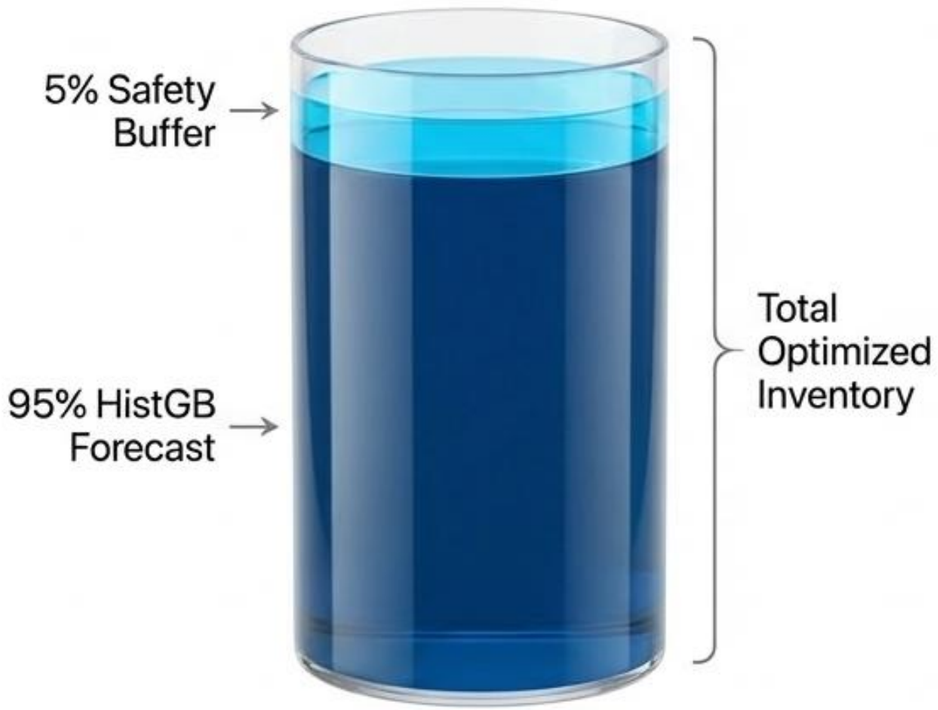
Why ML Wins:
Proactively capturing non-linear demand spikes.

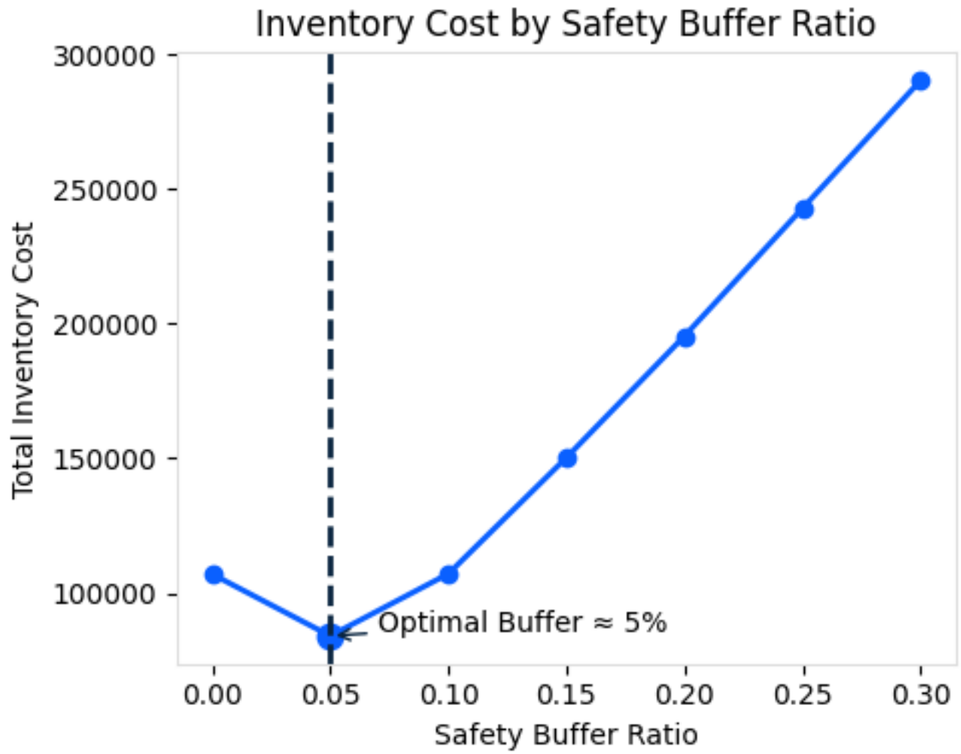


Demand spikes exhibit non-linear patterns.

While legacy models flatline during surges, feature-based ML proactively learns and captures this volatility.

Inventory Simulation:
The 5% safety buffer sweet spot





Simulation results show that a 5% buffer minimizes
the total inventory cost by balancing holding cost and stockout penalties.

Business Impact:
Redefining operational efficiency.

ML Precision (-21.8% Error) + Optimized Buffer (5%) = Maximum Capital Efficiency

Improving forecast accuracy is not just a statistical gain.
It enables smarter inventory decisions that reduce operational risk
while maximizing capital efficiency.